CALEB KUSUMO

System Architecture // Mechanical Design // 2026

ENGINEERING

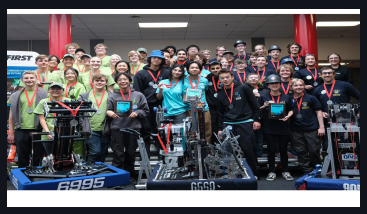
ENGINEERING PROFILE

Hi there! My name is Caleb Kusumo and I am a second-year Mechanical Engineering student at Boston University with a deep interest in applying the creative process to the intersection of mechatronics, industrial design, and manufacturing systems. Covering numerous areas, I have a strong background in robotics, electronics, manufacturing, and mechanical systems that stem from my experiences as the founder of a film business, CalebColor, where I primarily focus on camera repair and film sale, the captain and manufacturing lead of FIRST Robotics team 6560, and the steering lead for Terrier Motorsport at Boston University. Outside of my work, you can find me with camera in-hand or amidst a new project along the lines of those displayed here.



SYSTEM INDEX // [TOTAL UNITS: 9]

PROJECT ARCHIVE



2025 ROBOT

Ref_6560-25 / MECHATRONICS

Reefscape: Advanced vision-tracking and specialized intake for high-precision coral manipulation.

Solidworks / CNC / CAN Bus / Laser Cutting



2024 ROBOT

Ref_6560-24 / MECHATRONICS

Crescendo: High-speed swerve drivetrain with vision-aligned note delivery systems.

SolidWorks / Swerve / CNC

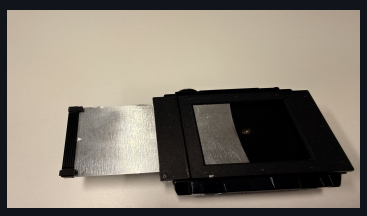


2023 ROBOT

Ref_6560-23 / MECHATRONICS

Charged Up: Features a piston-actuated arm and motorized gripper geometry capturing conical and square shapes.

Java / SolidWorks / Pneumatics / CNC

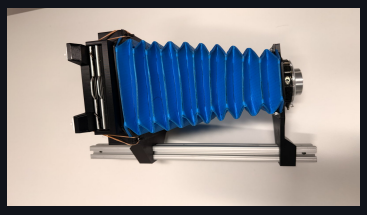


RB67 DARKSLIDE

Ref_CK-01 / MECHANICAL DESIGN

High-precision linear motion sub-assembly utilizing specialized coatings for low-friction operation.

SolidWorks / Waterjet / 3D Printing



4X5 RAIL CAMERA

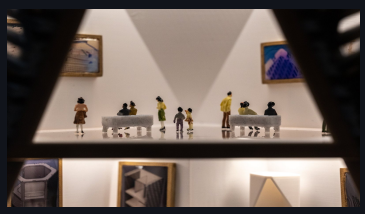
Ref_CK-04 / OPTOMECHANICAL DESIGN

Large-format modular camera system featuring 4-axis standard movements and high-extension bellows.

SolidWorks / 3D Printing / Camera / Laser Cutting

SYSTEM INDEX // [TOTAL UNITS: 9]

PROJECT ARCHIVE



LUTRON LIGHTING

Ref_CK-05 / ARCHITECTURAL SYSTEMS

Award-winning (3rd out of 40) lighting display for a 1:100 scale gallery space focusing on seamless design.

Fusion360 / Lighting Control / Circuits / Laser Cutting / 3D Printing / Water Jet



WAIST LEVEL FINDER

Ref_CK-06 / OPTOMECHANICAL DESIGN

A custom EF mount 1:1 ground glass waist-level finder, designed for through-the-lens viewing.

3d Printing / Optics / Solidworks



MAMIYA LENS HOOD

Ref_CK-07 / INDUSTRIAL DESIGN

A specialized 3D printed hood for the C330 TLR system, optimized for the 80mm f/2.8 lens.

3D Printing / Cameras / CAD / Solidworks



CAMERA REPAIR

Ref_CK-08 / MECHANICAL SERVICE

Shutter timing, film transport, light seal replacement, and lens service across a range of film and digital bodies.

Calibration / Repair / Optics / Cameras

EXPERIENCE

EXPERIENCE & BACKGROUND

EDUCATION

2025 - PRESENT - Boston, MA

B.S. MECHANICAL ENGINEERING - DEAN'S LIST (FALL '25)

BOSTON UNIVERSITY

2021 - 2025 - Irvine, CA

HIGH SCHOOL DIPLOMA - MAGNA CUM LAUDE, OUTSTANDING ENGINEER OF THE YEAR

UNIVERSITY HIGH SCHOOL

GPA 3.9 UW / 4.19 W

ENGINEERING EXPERIENCE

SEPT 2025 - PRESENT - Boston, MA

STEERING LEAD / AERO & BODYWORK TEAM MEMBER

TERRIER MOTORSPORT FORMULA SAE

Management of steering assembly design and fabrication. Collaborating on aerodynamic component fabrication and bodywork integration for formula-style racing vehicles. Applying advanced composite and manufacturing techniques to optimize vehicle performance.

SEPT 2022 - APRIL 2025 - Irvine, CA

TEAM CAPTAIN / MANUFACTURING & ASSEMBLY LEAD

FIRST ROBOTICS COMPETITION TEAM 6560 - CHARGING CHAMPIONS

Managed the design-to-final product timeline, leading a team to produce a competitive robot in under 4 weeks. Operated as Technician/Drive Team for REEFSCAPE and CRESCENDO, securing 2nd Place at Arizona North Regional, OC Regional Finalists, and Innovation in Control, Autonomous, and Creativity awards.

2024 - 2025 - Irvine, CA

MECHANICAL TEAM MEMBER

HYDROGEN GRAND PRIX

Assembled mechanical systems for a hydrogen-powered RC racer, focusing on endurance and fuel efficiency.

EXPERIENCE

EXPERIENCE & BACKGROUND

PROJECTS & ENTREPRENEURSHIP

FEB 2026

INDIVIDUAL COMPETITOR - THIRD PLACE

LUTRON LIGHTING COMPETITION

Designed, manufactured, wired, assembled, and presented a 1:100 scale lighting solution judged on creativity, engineering sophistication, and aesthetic appeal.

2023 - PRESENT

FOUNDER / LEAD TECHNICIAN

CALEBCOLOR FILM

Operated a high-volume film rolling and processing service, successfully developing 200+ rolls and selling 300+ rolls of film. Perform specialized mechanical repairs on film and digital cameras, including shutter timing, film advance, screen, sensor, and light seal replacement.

2022 - PRESENT

FREELANCE PHOTOGRAPHER

PHOTOGRAPHY

100+ photo sessions for seniors, model digitals, events, and weddings. Lead photographer for high-profile community events including Irvine Talks, OC South Asian Student Association, and Multi-Cultural Awareness Fashion Show. Managed post-production workflows for K4U Dance Club and Replay Performing Arts.

LEADERSHIP & VOLUNTEERISM

2025 - PRESENT

MENTOR - ELECTRICAL SYSTEMS, CNC, MILLING & LATHE

FIRST ROBOTICS COMPETITION TEAM 6560

Mentored junior members on electrical systems, CNC operation, milling, and lathe turning to ensure high-precision assembly.

2021 - 2025

STRUCTURES LEAD

STUDENT COUNCIL

Led and managed a team of 20+ members in construction of themed Spirit Night builds, including a life-sized Mystery Machine, Las Vegas sign, and gingerbread house.

TECHNICAL SKILLS

TECHNICAL SKILLS

DESIGN & SOFTWARE

SolidWorks
Fusion 360
Adobe Lightroom
Photoshop
Illustrator

MANUFACTURING

CNC Machining
Lathe & Milling
Waterjet
Laser Cutting
3D Printing (SLA/FDM)
TIG Welding
Vacuum Forming
Vinyl Cutting

ELECTRONICS

Soldering
Arduino
Raspberry Pi
Oscilloscope
General Circuitry

DEVELOPMENT

Python
C++
Java
Next.js

CONTACT

CONTACT ME

Available for technical consultation

DIRECT_COMM	NETWORK_LINKS
Email Me	**LinkedIn**
calebk@bu.edu	**GitHub**